

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
 ECE 552 Introduction to Computer Architecture
In-Class Exercise (10/31)

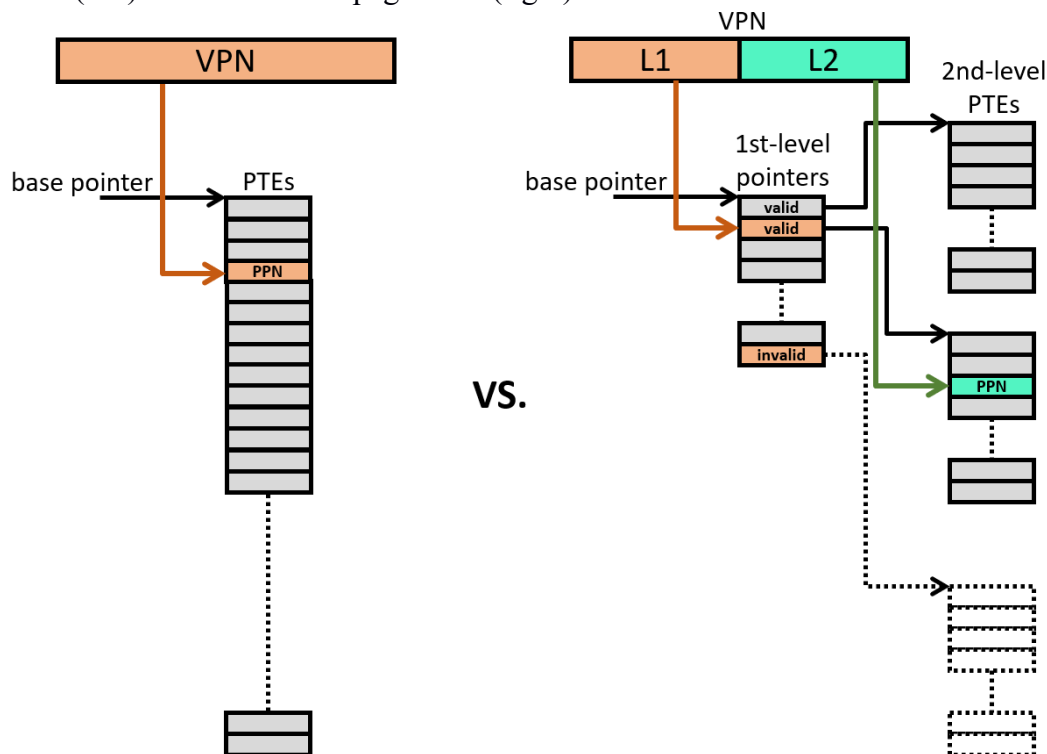
- You have a processor system with 32-bit virtual addresses and a **page size of 64KiB**; thus virtual page numbers (VPNs) are 16-bit. You have been running a program on this system, and it has accessed memory in the code, heap and stack segments as below. Assume all virtual addresses within the ranges shown are currently residing in physical memory (and no virtual pages outside these ranges have been allocated).

Code segment: 0x00000000 to 0x00001000

Heap segment: 0x20000000 to 0x320100A0

Stack segment: 0x7FFE00F0 to 0x7FFFFFFF

You are considering two different page table implementations shown below: a **one-level** page table (left) and a **two-level** page table (right).



- Consider the **one-level** page table implementation. If the program performs a load instruction for **virtual address** 0x2400005C, and after the page walk, the corresponding PTE contains the physical page number (PPN) 0x0123, what is the **physical address** (in hexadecimal) being accessed by this load?

Answer:

VPN 0x2400 was translated to PPN 0x0123. Now, concatenate {PPN,POFS} = {0x123,0x005c} = 0x0123005C.

- b) How many **bytes** of memory are needed to store the **one-level** page table implementation for the current program? Assume each page table entry is 4B.

Answer:

Number of entries in page table = 2^{16}

Thus $2^{16} \times 4\text{B} = 256\text{KiB}$

- c) Assume that the **two-level** page table implementation divides the 16-bit VPN into two fields: the most-significant 8 bits are used to index into the first-level table, and the least-significant 8 bits are used to index into the second-level tables. Assume that each entry in both tables is 4B. How many **bytes** of memory are needed to store the two-level page table implementation for the current program?

Answer:

VPN = {L1index, L2index}

Number of entries in first-level table = 2^8 (stored no matter what)

Number of entries in each second-level table = 2^8 (each L2 table stored only when L1 entry is not NULL)

Number of second-level tables = count of L1 indices { 0x00, 0x20, 0x21, ..., 0x32, 0x7F } = 21

Thus $(2^8 + 21 \times 2^8) \times 4\text{B} = 22,528\text{B}$

- d) You decide to add a **TLB** to this processor system. The TLB is **set-associative** and has a capacity of 128 PTEs. Assuming that the following **set index** and **tag bits** of the virtual address are used to access the TLB, fill in the blanks below.

virtual address		
31 : 20	19 : 16	15 : 0
tag	set index	page offset (POFS)
VPN		

The TLB is _____-way set-associative.

Answer:

$128 / 2^4 \text{ set index bits} = 8 \text{ ways}$

The TLB **reach** is _____ (in bytes).

Answer:

$128 \times 64\text{KiB page size} = 8\text{MB reach}$

2. Assume a processor system with a **one-level page table** and a one-level data cache hierarchy. Recall that the **page table resides in physical memory**, so a page walk requires one memory access. Suppose that the cache access delay is 2 cycles, and physical memory access delay is 100 cycles.

You run a program and find that the cache miss rate is 5% on average. Assume there are no page faults, PTEs are not cached in the data cache, and the cache is both indexed and tagged by physical address.

What would be the **average data access latency** (in cycles) for a load instruction:

- a) If this system were implemented **without a TLB**?

Answer:

$$\begin{aligned} &\text{page table access} + \text{cache} + \text{memory data access} \\ &= 100 + (2 + 5\% \times 100) \\ &= 107 \text{ cycles} \end{aligned}$$

- b) If this system were implemented **with a TLB**? Assume the TLB has an access delay of 1 cycle and a miss rate of 1%.

Answer:

$$\begin{aligned} &\text{TLB} + \text{page table access} + \text{cache} + \text{memory data access} \\ &= (1 + 1\% \times 100) + (2 + 5\% \times 100) \\ &= 9 \text{ cycles} \end{aligned}$$